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Managing design heritage - grandfather rights as a precursor for creative preservation

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ABSTRACT: This paper explores how creative preservation, affected by a regulatory framework, unfolds in the design of complex systems. Based on a case study of the Boeing 737 aircraft, it focuses on the role of grandfather rights, as part of the regulatory framework of aircraft design, as a precursor for creative preservation. The paper analyzes three design decisions related tot the evolving Boeing 737 aircraft models over a period of six decades and highlight the changing logic of creative preservation in relation to technology maturity, increasing complexity of design decisions, and expanded stakeholder involvement. Overall, the paper demonstrates that the management of design heritage is a 'living system' and that foundational practices may slowly become ineffective.

KEYWORDS: design management, decision making, design theory, design heritage, creative preservation

1. Introduction

The design of complex systems is not easily captured in one design approach or method as the character of these systems include engineering-intensive work, sophisticated organizational approaches, embeddedness in a larger network of actors as well as regulatory governance (Hobday, 1998; Lakemond et al., 2024). Their longevity and complex architecture create an important design heritage, that influences successive generations of the system (Barley et al., 2010). Sometimes, such design heritage is also referred to as legacy systems (Jaradat et al., 2022). In aircraft design, it is not unusual that successive aircraft rely on such legacy. For example, a considerable amount of Boeing's 737 cockpit design can be traced back to Boeing's 707, an aircraft with its first flight in 1954 (Brady, 2023). With the introduction of intelligent technologies, many complex systems are currently evolving into something that could be referred to as complex intelligent systems (CoIS) (Lakemond & Holmberg, 2022), i.e. systems with an emergent character, displaying inherent and recursive growth in diversity, scale and embeddedness (Yu et al., 2024). Yet, also CoIS still rely on an extensive heritage in terms of technology, subsystems, know-how, expertise, complex organizational processes, and networks. Future systems design will thus continue to rely on a delicate balance between managing the design heritage while simultaneously letting it evolve. This represents the nurturing of a logic that has been denoted as creative preservation (Hatchuel et al. 2019) and poses challenges for many providers of complex systems.
Consequently, there is an urgency to address how such a balance can be achieved, and how the logic of creative preservation unfolds. In this paper we focus on a particular aspect of this logic, i.e. the regulatory influence on design decisions and creative preservation. We aim to answer the following research question: How does creative preservation, affected by the regulatory framework, unfold in the design of complex systems? To find an answer on this question, we studied the extreme, but critical case (Flyvbjerg (2006) of the evolvement of one of Boeing's aircraft types, the Boeing 737, and how grandfather rights as part of the regulatory framework influenced important design decisions. The findings show that grandfather rights initially formed an important precursor for creative preservation and allowed for remarkable progress in the design of safe aviation, but successively obscured design decisions and reduced willingness to progress the design heritage allowing for new solutions. Eventually, grandfather

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rights impacted the possibility to determine the end of life of the complex system, not the least as technical design decisions were complex and increasingly relying on negotiations with a large network of actors introducing aspects beyond merely engineering design. Overall, the paper demonstrates that the management of design heritages is a 'living system' and that foundational practices may slowly become ineffective.

2. Theoretical framing

2.1. Design heritage in complex systems

Complex systems are recognized for their unique innovation and design characteristics, including their engineering intensity, reliance on a multitude of technologies, importance of system integration, many different actors involved, and regulatory involvement (Hobday, 1998). They tend to have an important role as the backbone of critical societal infrastructures, e.g. in transportation and energy. Examples include nuclear power plants, satellite systems, aircraft, road traffic management systems, and automated production systems.

The design of complex systems is well beyond single problem-solving activities and has been characterized by the indeterminate nature of design as well as the unique, complex, and wicked nature of design challenges (Hobday et al., 2012). Problems rarely have one single solution but are rather unique and complex and have multiple possible solutions (Buchanan, 1992).

Complex systems have been associated with a heritage that needs to be preserved while simultaneously something new is created (Appio et al., 2019; Lehtinen et al., 2019). This poses a tension between the new and existing (Calori, 2002), i.e. new solutions need to be integrated into an existing system architecture with existing solutions. Firms involved in the design of complex systems thus face a challenge of building on existing design, practices, standards, evolvement of existing technologies, while also integrating new technologies and resources (Bergek et al., 2013).

A design heritage is an important carrier of knowledge, representing knowledge structures that are expressed in technical solutions and design methods. Using heritage solutions allows for the creation of reliable systems building on proven technologies and may help to keep costs down (Barley, 2010). Yet, it also potentially constrains the introduction of new solutions and careful consideration that new solutions are not destabilizing the system itself (Hatchuel et al., 2019).

2.2. Creative preservation as a logic in design

Creative preservation has been put forward as one of contemporary society's most important design challenges (Le Masson et al., 2023). It presents a new logic beyond routine based design within a given knowledge system or innovative design which is considered as more creative and rule-breaking. It bridges not only existing and new technologies and solutions but also builds on a combination of innovation and tradition in work practices and organizations (Carvajal Perez et al., 2020). Such combination is full of dilemmas, e.g. related to fixations (Agogué et al., 2014) that counteract exploration activities, or to fixed ontologies that prevent revision of the identity of objects (Le Masson et al., 2023). In complex systems, the challenge of creative preservation is inherently embedded in design activities and related to the multitude of technologies, its longevity character and system integration activities. Strategies such as modularization and design rules (Brusoni & Prencipe, 2006) support creative preservation while at the same time enabling innovation. It has been suggested that the design process can explore and give new meaning to the existing rules (Harlé et al., 2021).

In industries with a high regulatory involvement, the design challenge related to a logic of creative preservation is further emphasized as regulations may constrain new innovative designs. Regulatory regimes, such as certification in aviation, put requirements on the design effort as well as the design organization and influence the possibilities to benefit from emerging technologies negatively (Oyeniya, 2018). However, they also have contributed largely to safety (Wolfe & NewMyer, 1985) and extremely reliable systems (Downer, 2023).

2.3. Creative preservation and safety in aviation

Aviation safety has reached levels of safety that are perceived as high by most. Safety continues to improve. During 2023, there were no fatal accidents with jet airliners, only one with a turboprop aircraft and in total 30 accidents. IATA summarizes this as *"At this level of safety, on average a person would*

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have to travel by air every day for 103,239 years to experience a fatal accident". Behind this impressive safety record is a meticulous system where all actors are certified and reviewed based on different aspects ranging from international and national governance, airports, air traffic management, airlines, pilots, maintenance operations. This extends also to supplying industries and their ecosystems.

In the development of aircraft, actors (OEMs and suppliers) must be approved by regulators as a design organisation, meaning that the actor is capable to develop and acquire aircraft, systems, materials and components that are airworthy, i.e. safe to fly, and of aviation grade. The approval verifies amongst others that a design authority is established by the actor through a design organisation, and that compliance/performance could be achieved and proved to achieve type certified products.

Early in the definition of a new aircraft design, the design organisation responsible for an aircraft negotiates the certification basis with the governing authority, the regulator, defining the basis to agree when the aircraft later is compliant and then obtaining the type certificate for the aircraft type. To ensure that an aircraft remains airworthy, a process of continued airworthiness is applied, including e.g. maintenance, logs and reporting systems. Aviation industry has continuously strived for improved safety, and the use of previous experience and knowledge plays a central role together with the ability to update systems when risks are identified, or when novel solutions may contribute to safety. A challenge is to benefit from previous demonstrations of compliance and allow the introduction of new solutions without overlooking potential unknown issues, while avoiding overly extensive tests and demonstrations of compliance.

The aviation industry has been able to master extreme safety in highly complex technology, building on what has been referred to as an astonishing depth and breadth of the myriad of intricacies of aircraft design and operation (Downer, 2023).

2.4. Grandfather rights

Grandfather rights have been part of aviation's regulatory framework for a long time. The concept of grandfather rights can be traced back to other contexts, initially representing a dark history where early applications were used to preserve racial inequalities in society (Sumers, 1914). Later, this practice expanded into the domain of food and drugs regulations, where something that was generally recognized as safe and effective was exempted from approval by corresponding government bodies (e.g. FDA in the US) (Nasr et al., 2011). A grandfathered drug did not allow for any change to the formulation, dosage form, potency, route of administration, indication/labeling, or intended patient population and needed to be fully documented.

Since roughly the 1950s, grandfather rights have been used in aviation industry with the purpose to enable further development of existing aircraft with new versions including improving efficiency, safety, stretches and modernizations. The arrangement of grandfather rights allows later derivatives of an initial aircraft type design to be manufactured under variations to the original Type Certificate of the aircraft. In this way, more complex procedures (such us a complete set flight tests) as part of getting approval for a new Type Certificate can be avoided.

By examining the specific phenomenon of grandfather rights as part of the regulatory framework and influencing creative preservation, we aim to shed light on the design challenges faced by organizations to manage a design heritage. There have been clear indications that there are some concerns related to the application of grandfather rights. For instance, the Joint Authorities Technical Review on B737MAX flight control system stated on 11 October 2019,

"Other broader recommendations raise the foundational issue of whether a process that has historically served the industry well for decades based largely upon compliance needs to be revisited to address not only compliance but also safety."

3. Method

To study how creative preservation, affected by the regulatory framework, unfolds in the design of complex systems we selected the case of the Boeing 737 (B737). B737MAX is highly relevant as a case to study design heritage as the aircraft represents an evolutionary derivative design. The aviation industry, characterized by its long-term stability amid technological changes, offers a fertile ground for studying creative preservation. The industry's institutional structure includes a range of stakeholders such as suppliers, customers, regulators, and professional bodies. These dynamics, coupled with competitive strategies centred around design and engineering, create a cauldron of interesting aspects. Specifically important in this paper is the regulatory framework related to certification and grandfather

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rights. These reflect the conditions for airworthiness, agreed early in the design of an aircraft, based on a design baseline that supports the negotiation of a certification basis. When developing new versions of aircraft, the certification basis is renegotiated, and it is during these negotiations the interpretation and application of grandfather rights are central to the design.

As the evolvement of B737 can be traced back to its first market entry 1967, and its latest model B737MAX has culminated into a frequently discussed case reflecting the limits of design heritage, it was deemed to be a good case illuminating the overarching research question. Given the events, it represents an extreme but critical case. Such a case has the potential to reveal more information, while informing on a more general problem (Flyvbjerg, 2006).

The research relies on an extensive array of secondary materials, including the official report and supplementary documents published by the House Committee on Transportation & Infrastructure in September 2020 investigating the design (Committee on Transportation and Infrastructure, 2020). This material comprises in-depth analyses and interviews with key informants. We have also used documentation from the extensive news coverage regarding B737MAX as well as blog pages for professionals in the industry. For the study, also extensive background material was used related to the design heritage of the B737 as well as the use of grandfather rights in general in the aviation industry. In the material, we have identified instances where grandfather rights were mentioned directly or indirectly and, for the sake of this paper, abstracted these from the material. The two authors have, based on individual and comparative joint analyses focusing on the role of grandfather rights as a precursor for creative preservation, resulting in the empirical description of the three central design decisions (see next section). To master the complex topic and material, we have relied on previous experience and relatively deep insights into the aviation industry. One of the authors has extensive industrial experience (almost 40 years) from this industry while both have performed previous studies in the aviation industry. Early findings have been validated in discussions with professionals in different types of roles in the aviation industry as well as through a network of people connected to the International Council of the Aeronautical Sciences.

4. Design decisions in the B737 case

The B737 is considered as one of the most successful commercial aircraft ever. As a narrowbody aircraft, it has been central to airlines for more than six decades and represented a substantial fraction of Boeing's turnover throughout these decades. The aircraft is still Boeing's offer in the narrowbody segment. This timespan must have been unimaginable for Boeing when the aircraft was designed in the early 1960s. An overview of the different generations is shown in figure 1.

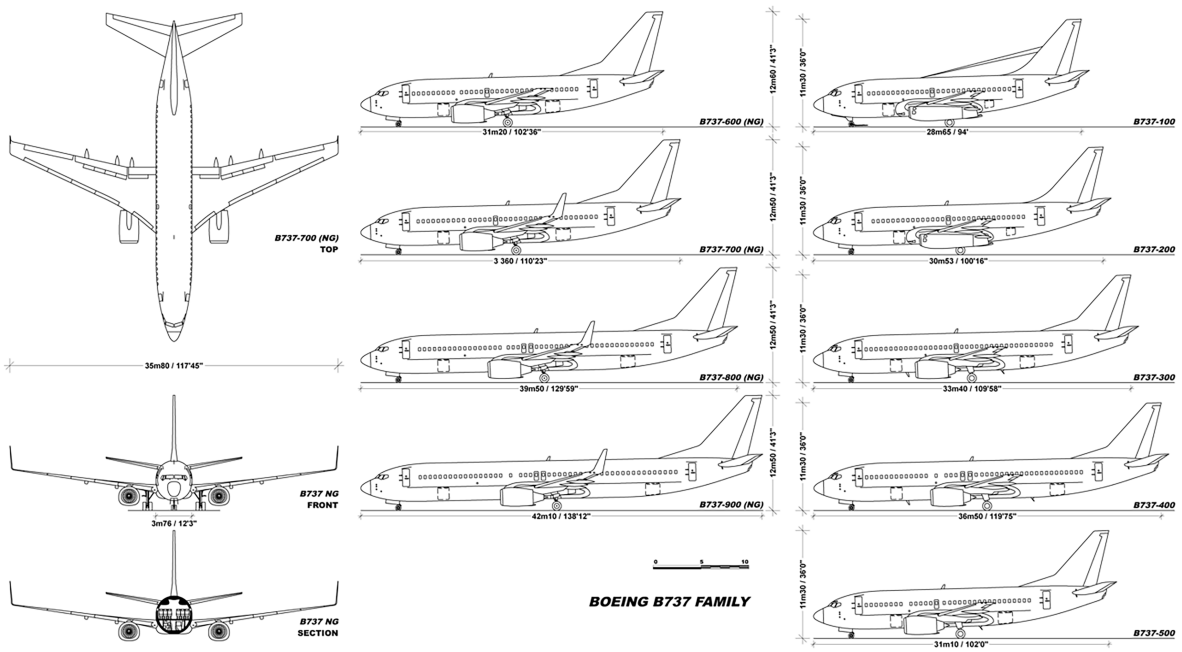


Figure 1. The B737 family (By Julien.scavini - own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=17180722)

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In relation to competitors, Boeing was late to the market with its B737. To enter the expanding market, Boeing identified a segment for its aircraft that would enable flying to smaller cities with small airports that had little infrastructure support at airfields and needed to keep operation costs down. When designing the B737, Boeing could rely on a design heritage including many wartimes designs as well as commercial aircraft, such as the B707 as the first US jetliner, and the B727 as the most manufactured jetliner of the time. Hence, as a well-established company for aircraft design, the design and configuration selection for B737 was a combination of using Boeing design experience and existing solutions for aircraft to keep time to market low while being able to bring in new technology and solutions that would offer new values compared to the competition.

To study design heritage of Boeings current B737 generation, we traced back the decision made connected to the development of the different generations since its first version in the 1960s.

In this paper we use B737 as a case to examine three fundamental design decisions further where the design heritage has played a significant role: (1) selection of the initial configuration with particular attention to engine integration and avionics in the early 1960s, (2) the upgrade to 737NG in the 1990s with attention to engine and wing upgrade and avionics relating to grandfather rights, and finally (3) the B737MAX upgrade in the early 2010s focusing on the selection of whether to make a derivative benefitting from grandfather rights, or a new model less tied to its design heritage.

4.1. Design decision 1 - selection of initial B737 configuration

The initial design configuration of the B737, the B737-100, addressed the mission to expand the market to enable flying to smaller cities and being efficient. It had a couple of defining design decisions. First, the aircraft was built low, and it carried stairs onboard the aircraft which reduced the need for airport infrastructure. Second, it relied on a low-bypass engine that was tightly integrated to the wing. This represented at the time an efficient configuration with limited drag and good structural efficiency that contributed to an efficient and cost-effective aircraft for operators. It was a configuration that none of the competitors had.

The initial design configuration provided limited room for fitting engines of larger dimensions. Interestingly, as Boeing manufactured turbines themselves until 1968, it can be expected that experts within Boeing knew that future engines would have an increasing diameter. This may have been considered and downplayed in the configuration decision, or maybe it was never salient in the decision process.

In addition, Boeing's urgence to reach the market, put high time pressure on the design and created an inclination to base the avionics solution largely on one of Boeings existing aircraft, the Boeing 727. By applying logics of similarity, Boeing could benefit from earlier designs and use the design heritage in aerodynamics, structures and systems. Such similarity logics is representing the process of using existing compliance documentation and data from a system or aircraft to demonstrate compliance for a nearly identical system or aircraft of equivalent design, construction, and installation. This is normal practice in the industry. For instance, it has been claimed that 60% of the cockpit design for Boeing 707 was kept in Boeing 727, and in turn 60% of that design was kept for the initial B737 (Brady, 2023). Even in B737NG there still exist components that originate from the Boeing 707 design.

4.2. Design decision 2 - the upgrade to 737NG in the 1990s

The initial version B737-100 was soon followed by B737-200 with the same engine integration. During early 1980s, the B737-300 to 500 were developed as derivatives with modernizations and larger diameter engines. To fit these engines on the original low aircraft, the engine covers are flat at the bottom to ensure enough ground clearance. This is a somewhat unique solution.

In the 1990s, competitive pressure from Airbus with the A320 was challenging the B737. Boeing was preparing for new versions of the B737 where upgrade possibilities included a new wing, new engines and an electronic flight control system. It was the wish of Boeing to include all three as a derivative of the B737 design. A derivative would cost less than the development of a completely new aircraft and the certification basis is different, where the older certification basis would contribute to making the aircraft more competitive. This led to extensive discussions not only between Boeing and FAA and other certification bodies but also engaging competitors of Boeing to voice objections. They posed that it would be unfair to consider such a radically different aircraft a derivative.

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The unfair advantage as perceived by Boeing's competitors can be exemplified by a much-debated issue concerning the rules for calculation of rejected take-off (i.e. aborting the take-off of an airplane after initiating the take-off roll but before the airplane leaves the ground) performance. These rules are defining a pilots reaction time, which in turn affect the allowed weight of the aircraft which then defines how much payload (passengers and freight) the aircraft can take. For the original B737, a lower reaction time was applied, as, during that time, pilots flew the planes *"almost like cowboys"* implying that they were very alert and could react quickly if for example an engine failed at a critical stage. Hence, in the regulations at the time, the reaction time for certification was set at one second. In the 1980s, the Airbus A320 as the main rival of the B737 was introduced based on a different philosophy, including a high degree of automation and fly-by-wire flight control system, the role of the pilot changed to more monitoring and programming the aircraft, rather than flying manually. Therefore, the pilot was not expected to be equally rapid to decide for a rejected take-off. Hence, the reaction time of the pilot was assumed to be longer in the case of a failing engine or another reason for aborted take-off. The regulation then stipulated two seconds reaction time for rejected take-off. This difference creates a competitive advantage corresponding to one more row of passengers for a normal flight if the aircraft could be certified as a derivative with the shorter reaction time of rejected take-off applied.

It was thus important for Boeing to get approval for a derivative design as this enabled benefitting from the certification basis, and, at least, reduced development costs by at least a factor two (Committee on Transportation and Infrastructure, 2020).

The outcome was the derivative B737NG, including acceptance for a new wing and new engines, but rejecting the full implementation of electronic flight control system. Consequently, the new generation still relied on an avionics architecture with high independence between the pilot and the copilot. The less integrated architecture did not support the full implementation of EICAS (Engine Information and Crew Alert System). In new Boeing aircraft types from the 1990s (e.g. Boeing 777 and later Boeing 787), such systems have been implemented, which may have influenced the possibilities for design decision 3. Still, B737NG represents important improvements in automation and cockpit technology supporting increased awareness support to pilots. It has been a step change in safety, improving safety records with a factor 10.

4.3. Design decision 3 - the B737MAX upgrade in the early 2010s

When arriving to the 2010s, it appears that Boeing had planned for replacing B737 with a new aircraft type. Internal studies on a replacement, called Yellowstone, had been going on since 2006. Also a new aircraft Boeing 787 was developed with a planned completion date before the B737 replacement offering opportunities to mature technology and system solutions before being implemented on a new narrowbody production line. Early in 2011, the Boeing's CEO communicated *"We're gonna do a new airplane. We're not done evaluating this whole situation yet, but our current bias is to not re-engine"*. Half a year later Boeing communicated that they instead were going to develop a derivative, the B737MAX. What made Boeing change their mind in such a short timeframe? Many aspects are of course involved in a discussion leading to that decision. Several aspects seem to have been central to the decision for a derivative; there was a large pressure from customers to develop a derivative; there was a strong confidence in the B737 and its ever-improving safety record; and the new Boeing 787 suffered numerous setbacks and heavy delays leading to less confidence in Boeings ability to develop a new aircraft on time. These aspects together with the higher cost for a new aircraft and Airbus launching A320Neo in a timeframe that would make a new aircraft late to the market, created an enormous pressure for Boeing to come up with a competitive derivative that also had to have such similarity in behaviour that simulator training would not be necessary for pilots certified for B737NG.

At this time, the design heritage started to create some severe challenges. To attain a competitive efficiency of the aircraft, it was key to fit an engine with an as large as possible diameter under the wing. This, together with increased power etcetera, led to slightly changing handling qualities when flying causing such a difference for pilots that simulator training would be needed. Avoiding this, in an aircraft that was not fly by wire, was not straightforward. The solution was to introduce a system called MCAS. This system was however not easy to integrate and support due to the less integrated architecture. Still, it was not safety-critical, as it had such low authority that pilots would be able to override it, and it was also understood to hardly ever be used. However, during testing, it became clear that the control

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characteristics had larger issues than expected. This was resolved by increasing the authority of the MCAS system, making the system safety-critical, something that was not catered for. A following number of mistakes consumed safety margins, and this eventually caused two fatal accidents during the early years of operation and leading to grounding of the aircraft for a long time (i.e. judging the aircraft type as not airworthy). Please note that our simplified discussion has no intention to describe the exact causes and responsibilities but rather put forward the issues in managing a design heritage, creative preservation and the role of grandfather rights in design decisions.

Design decision 3 includes aspects such as short-term vs long-term considerations, emerging issues, conflicting internal design goals, external tensions based on amongst others market preferences for a derivative and low conversion costs, and tensions between regulators/regulations and preferred design solutions partly originating in grandfather rights. After the accidents, it was expressed in the report investigating the design, development and certifications of B737MAX (Committee on Transportation and Infrastructure, 2020) that *"The MAX crashes were not the result of a singular failure, technical mistake, or mismanaged event. They were the horrific culmination of a series of faulty technical assumptions by Boeing's engineers, a lack of transparency on the part of Boeing's management, and grossly insufficient oversight by the FAA"* and in the senate hearing by Sully Sullenberger (the Hudson miracle pilot) that is quoted in the report *"... Accidents are the end result of a causal chain of events, and in the case of the B737 MAX, the chain began with decisions that had been made years before, to update a half-century-old design..."*. Both express the importance of managing design heritage.

5. Analysis and discussion

To unravel how creative preservation, affected by the regulatory framework, unfolds in the design of complex systems, especially related to the role of grandfather rights as a precursor for creative preservation in B737, the discussion centers around design decision 3, and relates back to consequences of design decisions 1 and 2. All three decisions concern a delicate balance of when to use existing solutions and design knowledge and when to use novel and less well-known solutions. This is a well-known dilemma in complex systems (Hobday et.al. 2012; Appio et al., 2019; Lehtinen et al., 2019), and in aviation particularly related to progressing a system towards better performance and safety (Downer 2023).

The rationale of grandfather rights in aviation is to combine the benefits of a rigorous certification with a resource and cost-efficient implementation of innovations contributing to safety and performance. Obviously, the practice has been successful, resulting in highly reliable aircraft and drastically improved safety records over the years.

In the early years of applying grandfather rights in aviation, the findings indicate that it was relatively straightforward to design improved aircraft and gaining in safety and performance, by relying on a certification process with grandfather rights. Over time however, as aircraft have become more mature, complex, technologically advanced, and safety has progressed by an order of magnitude, the applications of grandfather rights have centred increasingly around decisions of *"what constitutes a new type of aircraft vs. a derivative"*. This reflects an increasingly complex situation to understand safety implications.

The insights from the case have several implications for understanding how creative preservation, affected by a regulatory framework evolves. First, based on the findings, it is proposed that the level of technological maturity and understanding of the consequences of a change are impacting the willingness to progress the design heritage and allowing for new solutions. This is reflected in design decision 2, where the applications of grandfather rights seemingly favoured mature technology (cf. Oyeniya, 2018). As involved parties had a mature knowledge and ability to analyse the consequences, a new and more efficient wing was accepted within the type. In contrast, at the time, a fly by wire system was relatively novel and more difficult to master. Hence, a fly by wire system was not accepted within the type. This reflects difficulties in balancing creativity and preservation in the creative preservations logic and puts an emphasis on the intricacies of dealing with a non-salient or a less well-known future, while this amplifies future difficulties to deal with a design heritage (cf. Carvajal Perez et al., 2020). In our case, acceptance of a fly by wire system in design decision 2, would have enabled a design decision 3 that probably would have required less focus on dealing with constraints or impossibilities due to grandfather rights. It would have progressed the design heritage and its relevance extending into the future and perhaps extending the life of the product.

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A second important insight relates to the indeterminate nature of design (Hobday et al., 2012), where the logic of creative preservation is not easily made salient when cause and effect relationships cannot be easily understood and traced back within a design heritage. In addition, it raises the attention to whom is involved in the design process and decisions related to creative preservation. In our case, certification is normally negotiated between a regulator and a responsible design organisation. However, this expanded as discussions on the interpretation of grandfather rights became central to competition and to customers. It certainly complicated the negotiation. For example, in design decision 3, customers took a position that they would prefer a derivative, and competition tried to lock-in Boeing in a decision for a derivative by launching a derivative themselves. This pushed Boeing to stretch what was possible within the design space of a derivative design to remain competitive.

A third insight therefore extends to the influence of design decisions relating to creative preservation and the design heritage on the available design space. Both design decision 1 and 2 had strong implications on what was possible with a derivative, influencing and constraining the available design space for designing a competitive product.

Fourth, it appears that the role of grandfather rights creative preservation has transformed over time, from a precursor to a new meaning (cf. Harlé et al., 2021). From being early on an efficient tool to develop safer aircraft relying on certification principles, grandfather rights increasingly evolved towards involving an expanding set of stakeholders, calling for design efforts that have an origin in constraints imposed by the previous application of grandfather rights, and perhaps increasingly obscuring the understanding of would be the best way forward in combining design heritage and novel solutions.

The transformation of the role of grandfather rights seems to have impacted the possibility to determine the end of life of the complex system. It appears that stakeholders had difficulties in agreeing on whether the product was near end of life or not. Grandfather rights had evolved from their original focus on safety. The more tactical use to achieve competitive advantage clearly favoured a derivative, while a stricter safety focus would have pointed towards declaring end-of-life. Tragically, design decision 3 resulted in a clear and unexpected drop in safety of the B737 aircraft. In this case, it could be posed that the negotiations in design decision 3 and grandfather rights application have negatively contributed, while potentially grandfather rights have served and could serve well to better understand the bounds of continuing further with an existing design.

6. Conclusions and limitations

The exploration of how creative preservation, affected by the regulatory framework, unfolds in the design of complex systems, provides several contributions illuminating aspects of and insight into the logic of creative preservation and the management of design heritage in the design of complex systems. First, grandfather rights have been effective and serving the industry well by enabling continuous development of aircraft benefitting from design heritage and innovation leading to safer, more reliable and efficient aircraft as a collective practice incorporating the individual interest of the different actors.

Previously, a relatively limited number of stakeholders, i.e. industry, design authority and the governing authorities (e.g. EASA and FAA), have been able to successfully oversee, balance and manage aspects of design heritage. However, as aircraft have become more complex, e.g. with the integration of increasingly software-based functionality (Lakemond et al., 2024) and electrification, this has become more difficult. Therefore, a second contribution is related to complexity and the number of different stakeholders, altering the conditions for achieving effectiveness in collective practices. In addition, the role of grandfather rights has expanded to negotiations not only between industry and authorities, but also been central to e.g. market issues.

A third contribution is thus connected to the expanded focus of negotiations, beyond merely design considerations and making negotiations increasingly complex and difficult to navigate as stakeholders have a range of intentions obscuring the initial goals of enabling best practice of aircraft development. These contributions point at a need to alter the foundational practices of grandfather rights as a precursor for creative preservation. Long-term, apparently the management of design heritages is a 'living system' and that foundational practices that have been effective may slowly become ineffective. Here, as in aircraft design, the capability to be able to identify when to preserve and when to be creative is central. This entails both product and design practices.

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The paper reports on the intricate task to manage a design heritage and explores the role of grandfather rights as part of a regulatory framework governing the design of complex systems. While we build the paper on extensive evidence, it is almost inconceivable to grasp the details of all design decisions that have been made over time. As an extreme case the findings may not be representative for all other types of industries. However, we align with Flyvbjerg (2006) who has posed that extreme cases can say more about a phenomenon than typical cases. They have a potential to cultivate more reflexivity and allow individuals and collectives to mindfully consider organizing possibilities and trade-offs (Chen, 2015). Hopefully, our exploration will inspire researchers and practitioners to continue exploring how organizations involved in complex systems can reap the benefits of their design heritage while progressing design to contribute to a fruitful future.

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